Topic: Binary-search confidence and bubble sort

WAGBA: Follow how an algorithm changes its data, and explain why each step happens.

Keywords: index, value, target, low, mid, high, adjacent, swap, pass, flag.

MAIN TASK 1 | GUIDED BINARY SEARCH | 8-20 MINUTES

Read, point, compare, record.

Name: _____ Class: _____ Date: _____

This booklet is for Main Tasks 1 and 2 only (Q2-Q7). Complete Q1, Types of Learning, Q8, the Learning Pit Stop and Q9 in the web app. Keep the original question numbers to match the screens.

Pseudocode: read and trace on paper; do not paste it into IDLE. The symbol <- means store the result in. Python is a different, runnable representation.

Routine: mark low/high indexes; calculate mid; read its value; compare; move one boundary or stop; record. Keep the original indexes.

Index	0	1	2	3	4	5	6
Value	**5**	**9**	**13**	**18**	**24**	**31**	**42**

Q2. Follow the screen example searching for target value 13. Complete each row before the next decision is revealed. Initial low = 0; high = 6. This is guided practice, not an independent score.

Midpoint index = (low + high) DIV 2. DIV is integer division, so 5 DIV 2 = 2.

Iteration	Low index	High index	Mid index	Value at mid	New boundary / found
1					
2					
3					

If target < midpoint value: high <- mid - 1.
If target > midpoint value: low <- mid + 1.
If they are equal: found <- True; stop.

Q2d. Why exclude the checked midpoint when it is not a match?

A smaller next step is allowed: point to the remaining positions before writing the new index.

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MAIN TASK 1 | INDEPENDENT CHECK

Try a fresh search.

Index	0	1	2	3	4	5	6
Value	**4**	**8**	**12**	**19**	**26**	**33**	**40**

Q3a. Search for target value 26. Initial low index = 0; high index = 6. Record each iteration and the boundary change or found position. [6]

Iteration	Low index	High index	Mid index	Value at mid	New boundary / found
1					
2					
3					

Q3b. Which indexes did your first decision rule out? Explain why using the target value, midpoint value and sorted order. [2]

Q3c. Why should low store an index rather than the target value 26? [1]

Optional challenge while your teacher supports others

Using the same list, search for target value 1. List the midpoint indexes checked. Explain how low > high shows that the target is absent.

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MAIN TASK 2 | FACTS 1 AND 2 | 20-30 MINUTES

Compare neighbouring values.

Bubble sort compares adjacent values, swapping them if the left value is greater. Ascending order means smallest to largest. A pass checks every adjacent pair once.

Index	0	1	2	3
Value	**7**	**3**	**5**	**2**

Q4a. If i = 0, which values does numbers[i] > numbers[i + 1] compare? Should they swap? Write the whole list after that comparison. [2]

Q4b. Follow the app through the first complete pass. Record the whole list after each comparison. Each row starts from your previous row, not the original list. [3]

Comparison	Indexes compared	Whole list after this comparison
1	0 and 1	
2	1 and 2	
3	2 and 3	

Check your last row: this is the list after one pass, not necessarily the final sorted list.

Before moving on, point to the value that reached its final position. You do not need an additional written answer.

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MAIN TASK 2 | FACT 3 AND PASS TRACE | 30-50 MINUTES

Repeat a pass, or stop?

Q5a. Why is one correctly ordered pair not enough to stop bubble sort? What must a whole pass show? [2]

Q5b. Start again with [7, 3, 5, 2]. This version compares all three adjacent pairs on every pass. Complete the table, including the final no-swap pass. [4]

Complete pass	Whole list at the end of this pass	Any swaps? True / False
1		
2		
3		
4		

Q5c. Why set swapped = False at the beginning of each pass? [1]

Words you can use

every adjacent pair | ascending order | complete pass | no swaps | reset | this pass

If stuck, show the last row you are confident about. Continue from that row; do not restart the whole task.

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MAIN TASK 2 | Q6

Complete the Python on paper.

Q6a. Write the missing Python for gaps A-D below. This is Python, but you do not need IDLE or an online editor today. [4]

```
 1  numbers = [7, 3, 5, 2]
 2  swapped = True
 3  while swapped:
 4      swapped = ____ A ____
 5      for i in range(____ B ____):
 6          if numbers[i] ____ C ____ numbers[i + 1]:
 7              temp = numbers[i]
 8              numbers[i] = numbers[i + 1]
 9              numbers[i + 1] = temp
10              swapped = ____ D ____
11  print(numbers)
```

A: _____ B: _____

C: _____ D: _____

Q6b. Why must i stop at len(numbers) - 2? Explain using i + 1 and the last valid index. [2]

Read the swap carefully

Line 7 saves the old left value in temp. Line 8 copies the right value left. Line 9 restores the saved value on the right. Both values are preserved.

Remember: range excludes its stop value. For four items, range(3) produces i = 0, 1, 2.

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MAIN TASK 2 | Q7 | 30-50 MINUTES

Use test evidence, not a guess.

Q7. Hand-trace the completed Q6 program for each test list. Record the final printed list and the number of complete passes, including the final no-swap pass. [6]

Initial test list	Final printed list	Number of passes
[7, 3, 5, 2]		
[2, 3, 5, 7]		
[5, 2, 5, 3]		

Use this space for intermediate lists or comparisons. This is predicted paper evidence, not an actual program run.

Finished Main Task 2? Keep this booklet beside you and return to the web app for Q8 (extension), the Learning Pit Stop and Q9 (plenary). Those responses belong on screen, not here.

Submission: hand this booklet to your teacher. Export your on-screen answers separately using the app's PDF button, then attach that PDF to the Teams lesson assignment and select Turn in.